REQUIREMENT ANALYSIS REPORT

Project Title

E-Commerce Order Management Database System

Introduction

The E-Commerce Order Management Database System is designed to manage the complete online shopping process efficiently. The system maintains customer details, product information, supplier records, orders, payments, shipments, and customer reviews in a centralized database. It helps the organization improve data accuracy, reduce redundancy, and enhance operational efficiency.

Objective

The main objective of this project is to analyze the business requirements of an E-Commerce Order Management System and identify the necessary database entities, business processes, and system requirements. This analysis serves as the foundation for designing an efficient and reliable database system.

Business Scenario

A rapidly growing e-commerce company requires a centralized database to manage its daily business operations. The company needs a system that supports customer registration, product management, inventory tracking, order processing, payment management, shipment tracking, and customer reviews. The proposed database system will improve data consistency, increase operational efficiency, and support business decision-making through accurate information.

Core Entities Analysis

The system consists of the following core entities:

Customer

Stores customer details such as Customer ID, Name, Email, Mobile Number, Address, and Registration Date.

Category

Stores product categories including Category ID, Category Name, and Description.

Product

Contains Product ID, Product Name, Price, Stock Quantity, and Category ID.

Supplier

Maintains supplier information including Supplier ID, Supplier Name, and Contact Information.

Order

Stores Order ID, Customer ID, Order Date, Total Amount, and Order Status.

Order Details

Stores Order Detail ID, Order ID, Product ID, Quantity, and Unit Price.

Payment

Maintains Payment ID, Order ID, Payment Method, Payment Date, and Payment Status.

Shipment

Stores Shipment ID, Order ID, Delivery Address, Shipment Date, and Delivery Status.

Review

Stores Review ID, Customer ID, Product ID, Rating, and Comments.

Business Processes

The major business processes are:

- Customer Registration
- Product Management
- Category Management
- Supplier Management
- Order Placement
- Payment Processing
- Shipment Tracking
- Customer Review Management

Stakeholders

Customer, Admin, Supplier, Delivery Staff, Payment Gateway.

Functional Requirements

• Customer registration

• Product management

• Order placement

• Payment processing

• Shipment tracking

• Review management

Functional Requirements

The system should:

1. Register new customers.
2. Maintain customer records.
3. Manage product information.

4. Manage product categories.
5. Maintain supplier details.
6. Process customer orders.
7. Store order details.
8. Process payments.
9. Track shipment status.
10. Allow customers to submit reviews.
11. Generate business reports.

Non-Functional Requirements

The system should provide:

- High Security
- Fast Performance
- Data Accuracy
- Reliability
- Availability
- Scalability
- Easy-to-use Interface
- Data Backup and Recovery

Project Scope

The project covers:

- Customer Management
- Product Management
- Category Management
- Supplier Management
- Order Management
- Payment Management
- Shipment Management
- Review Management
- Report Generation

Assumptions & Constraints

Assumptions

- Customers have internet access.
- Payment gateway is available.
- All customer information is valid.
- Products are updated regularly.

Constraints

- Only the given core entities are used.
- The database is centralized.
- Internet connection is required.
- Users must have proper authentication.

Conclusion

The Requirement Analysis Report provides a clear understanding of the business requirements for the E-Commerce Order Management Database System. It identifies the stakeholders, core entities, business processes, functional requirements, and non-functional requirements. This analysis forms the basis for the database design and ensures that the system meets the organization's operational needs effectively.